Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 · Electrochemistry and corrosion · Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II

1 Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II

Study focus: Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000142 · EV0000143

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module six on corrosion chemistry and we will talk about different types of electrochemical corrosion and today's discussion will be on waterline corrosion and petting corrosion.

Key details

- Last class we discussed about differential aeration corrosion where an oxygen concentration cell was getting formed due to difference in the concentration of oxygen to which the metal surface is exposed.
- This happens when a metal is half immersed in water.
- And when this happens we call it as waterline corrosion.
- Whereas at the cathode because there is more oxygen.
- And if you remember electrochemical theory of corrosion, I said that the.
- Products which are formed at the anode and cathode, the ions which are formed at the anode and cathode move towards each other and in between the anode and cathode we get the formation of the corrosion product.
- The corrosion product will deposit in between the anode and cathode.
- And that's exactly what happens here.
- A brown line on the surface of the ship.

- Many of you would have noticed this, or there will be a brown line on the surface of the ship, and where do you see it?
- Wherever the water line.
- Brown line and what is that brown line on the surface?
- It is due to the corrosion product.
- they move towards each other and somewhere in between they will get deposited and that is happening at the water level.
- Which is formed just below the waterline, and because of this, this type of corrosion is referred to as waterline corrosion.
- So basically it is differential aeration, but because it is happening in the presence of water, especially, it is called waterline correction.
- Corrosion and the brown line appears at the waterline.
- What happens here?
- The part which is inside the inside the water now here again exposed to less oxygen that becomes the anode and the part which is above exposed to more oxygen is going to become the cathodic area and in between we will see a brown line getting.

Example or application from the lecture

- Let us see, take up some examples and understand this.
- Now let us take up another example where we have a steel tank which is partially filled with water and left for a long time.

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Core explanation · 05:00–09:59

formed on the surface of this steel tank.

Key details

- So half of it is inside the water, again the same reason we have less oxygen, therefore iodic area is this oxy this part.
- And the more oxygen part would be the cathodic area, and somewhere at the waterline we will see the formation of the corrosion product.
- And now, since there is sodium chloride, we know that.
- The conductivity of the solution will be very high, and therefore we find corrosion also will be more in this case when we have a solution.
- Of course, it is aerated, so obviously there will be oxygen in it, but never.
- the less oxygen.
- What happens here?
- It is a localized and accelerated corrosion, and because of this, there is formation of pits and pin holes around which the metal is relatively unattacked.
- You can see in this picture how pits have got formed on the metal surface, and it's only this area which has got corroded, whereas the rest of it looks good.
- And therefore we find that formation of pits and pinholes.

- Therefore it is called pitting corrosion and the rest of the area of the metal looks intact.
- Because if there are very small pinholes getting formed, the corrosion product itself is going to cover the pinholes and we will not even know till the destruction has happened.
- Which is carrying oil or water.
- And the pipe will suddenly start till it is completely damaged, we will not come to know and unless it starts leaking, it will just give out.
- the water or the oil and that is where that is when we come to know but the damage is already done.
- So what is the characteristic very important here for us to understand it is characterized by small anode area and a large cathode area because the anode area is very small it will result in accelerated corrosion.
- Now, because the water droplets is covering a small part of the metal surface, the entire surface is now not exposed to the atmosphere, a small part.
- That is covered by this water droplet.
- As a result, what is going to happen?
- You can see there is a small anode and a large cathode, as I said just now that a small anode and large cathode is characteristic of fitting corrosion, and we will find that the anode.

Example or application from the lecture

- Another example would be when an iron rod is partially dipped in a sodium chloride solution.
- We have to understand that waterline corrosion is also differential aeration corrosion, but only a special case because we consider these cases wherever the metal is partially immersed into the water.
- For example, if this is a pipeline.

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Core explanation · 10:00–14:59

Anode being small undergoes corrosion at a very fast rate.

Key details

- You can see again here there's a small anode and there's a large cathode which is there.
- Now because the area the cathode is so large, the demand for electrons is very high.
- from the cathode.
- The anode will undergo corrosion at a faster rate.
- And because the anode is very small, we will find that it undergoes corrosion at a very fast rate being small there is like a.
- Which is getting formed in this anode area.
- Why accelerated corrosion?
- We can understand because the demand for electrons is very high, the cathode being large.
- Taking part in the reaction, of course, here because it is exposed to more oxygen, we will say this is the reaction which is happening under the dust particle or the water particle, it is

anode, so it is the metal.
- Say we were talking of iron or steel, so iron will become Fe2 plus.
- There is destruction, and this pit will go on deepening slowly till it reaches the end.
- And till that time we will not be able to even discover that there is a pit, and that will lead to a damage of the structure.
- could be of fitting corrosion where we have this protective film which got or formed on the surface which is be been made on the surface has got damaged.
- This process is calling.
- But the moment there is a damage or there's a small hole or it got scraped, then this is going to expose the iron metal to the surface.
- And this is minus zero point four.
- These are the electrode potential values.
- Iron is going to act as the anode and tin is going to act as the cathode.
- So this now will result in the anode undergoing corrosion at a very fast rate and therefore since the anode is very small it will be like digging a pit in this area.
- So therefore, we find that in this case tin is acting as the cathode.
- The value is minus 0.18 or maybe 0.14.
- And here it is 0.44, and therefore we find that this metal.
- The steel is going to act as anode, whereas the tin is going to act as cathode, and now anode being very small, the demand for electrons from the cathode, the tin coating is the cathode large cathode, demand for electrons is very high, and therefore it forces the anode.
- To undergo corrosion at a very fast rate, and because the anode is small, it will be a pit will be dug here, and then this is how the damage is going to happen.
- Or we say galvanic corrosion, whereas in the case one we saw it was differential aeration corrosion.
- But what is common to go?
- That they have a small anode and a large cathode, and that is what is characteristic of pitting corrosion.

Example or application from the lecture

- Let us understand this better.
- For example, this is a very common thing which we see iron being covered by tin.

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Core explanation · 15:00–19:59

Tin is not going to be involved in the reduction reaction, it is only the environment, and here we have oxygen.

Key details

- So therefore this is the reaction which will take place.
- Hence the anode will get corroded and a deep pit, narrow pit will be formed at this position.
- What we have to remember small anode and large cathode.

- Whenever we have this situation, it is going to result in pitting corrosion.
- So before that let me tell you that in the next class we are going to discuss the last type of corrosion that is stress corrosion.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9963; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

C Revision and retrieval practice